

NBA Head-to-Head Win Probability Prediction

Team: Jason Qin, Marshall Scott, Pengfei Zhang
Project Mentor: Alec Traaseth

Data Science Boot Camp, Erdos Institute, Summer 2026

Introduction

- **Context**

- Sports betting is as old as sports
- With the rise of FanDuel, DraftKings, Kalshi, and Polymarket, sports betting is more popular than ever
- Basketball is the third most popular sport in America and widely speculated on

- **Question**

- Can the winner of an NBA game be predicted?

- **Goal**

- Develop a classification model that can predict the probability of the home team winning a head-to-head match with an away team



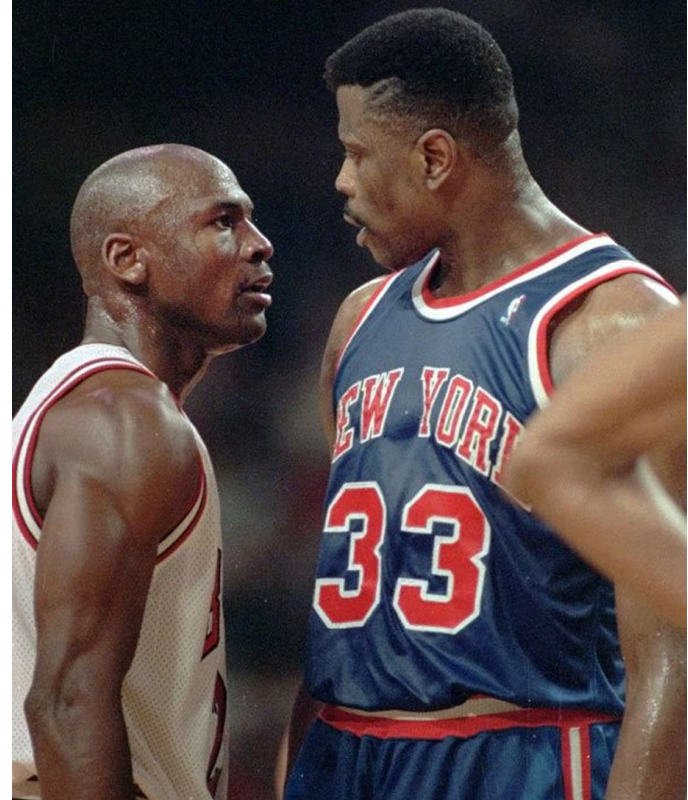
Introduction

- **KPIs**

- Prediction Accuracy
- ROC-AUC

- **Stakeholders**

- Prediction markets
- Sports betters
- General managers
- Coaches



Data Cleaning

Source: Kaggle NBA Box Scores & Stats
(1947–today) by Eoin Moore

Restricted to 2010+ (games evolve quickly)

Removed noncompetitive games (preseason
and all-star matchups)

20,518 games → 19,147 regular season +
1,371 playoff

	gameId	gameDateTimeEst	teamId	teamName	opponentTeamId	opponentTeamName	home	win
0	42500405	2026-06-13 20:30:00	1610612759	Spurs	1610612752	Knicks	1	0
1	42500405	2026-06-13 20:30:00	1610612752	Knicks	1610612759	Spurs	0	1
2	42500404	2026-06-10 20:30:00	1610612759	Spurs	1610612752	Knicks	0	0
3	42500404	2026-06-10 20:30:00	1610612752	Knicks	1610612759	Spurs	1	1
4	42500403	2026-06-08 20:30:00	1610612759	Spurs	1610612752	Knicks	0	1
...	...	...	...	...	...	...	...	...
43541	11000004	2010-10-05 19:00:00	1610612739	Cavaliers	1610612766	Bobcats	1	1
43542	11000007	2010-10-05 19:00:00	1610612751	Nets	1610612755	76ers	0	1
43543	11000007	2010-10-05 19:00:00	1610612755	76ers	1610612751	Nets	1	0
43544	11000003	2010-10-04 15:00:00	1610612750	Timberwolves	1610612747	Lakers	1	1
43545	11000003	2010-10-04 15:00:00	1610612747	Lakers	1610612750	Timberwolves	0	0

Feature Selection and Engineering

Two ways to encode past performance:
10-game rolling average, exponential
weighted mean

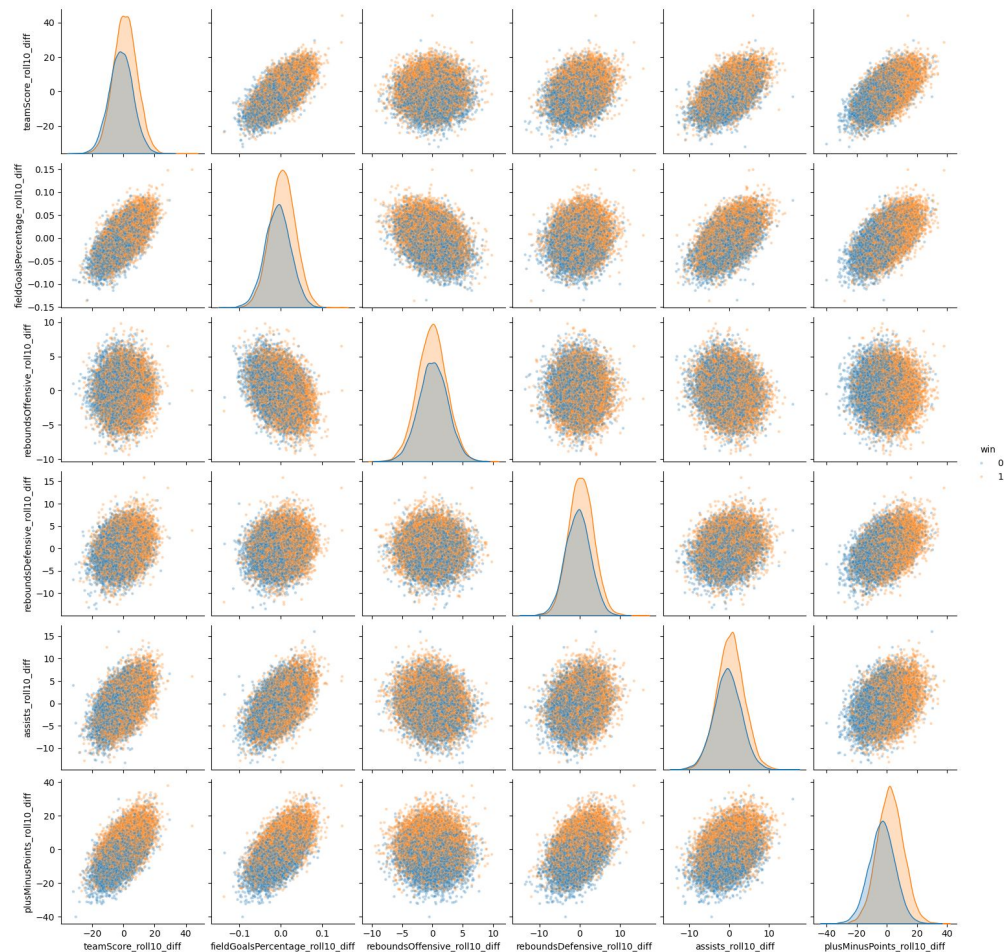
Leakage prevention: `shift(1)` before
computing any rolling/EWM stat

Context features: rest days,
back-to-backs

Features built per team **and** as
home-away differentials

	gameDate	home_teamName	away_teamName	win	teamScore_roll10_diff	fieldGoalsPercentage_roll10_diff	plusMinusPoints_roll10_diff
0	2010-10-30	Hawks	Wizards	1	28.500000	0.147500	38.000000
1	2010-11-03	Hawks	Pistons	1	10.750000	0.038750	17.750000
2	2010-11-07	Hawks	Suns	0	-3.366667	0.037700	9.433333
3	2010-11-10	Hawks	Bucks	0	12.250000	0.085000	5.250000
4	2010-11-12	Hawks	Jazz	0	-1.694444	0.036042	2.236111
...	...	...	...	...	...	...	...
20513	2026-03-29	Hornets	Celtics	0	5.800000	0.016900	5.600000
20514	2026-04-02	Hornets	Suns	1	3.200000	0.020700	9.800000
20515	2026-04-03	Hornets	Pacers	1	-1.700000	-0.040500	18.500000
20516	2026-04-10	Hornets	Pistons	0	-0.800000	-0.035400	4.400000
20517	2026-04-14	Hornets	Heat	1	-12.400000	-0.044800	6.700000

EDA: win/loss with historical data



Exploratory data analysis (EDA)

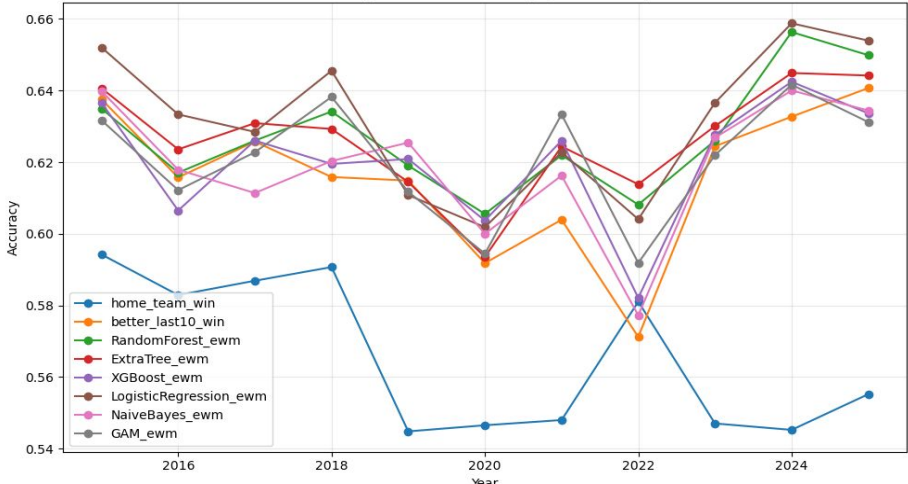
Pairwise plots of selected features against each other of home team wins vs losses.

Weak, but visible correlations between the performance features and the target (home team win)

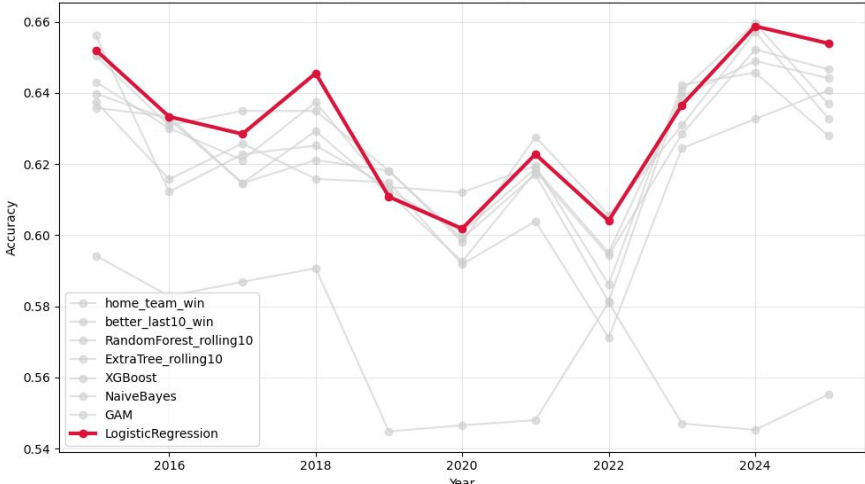
Model Comparison

	home_team_win	better_last10_win	RandomForest_rolling10	ExtraTree_rolling10	XGBoost	LogisticRegression	NaiveBayes	GAM
year								
2015	0.5942	0.6375	0.6431	0.6504	0.6358	0.6520	0.6561	0.6398
2016	0.5829	0.6157	0.6301	0.6309	0.6333	0.6333	0.6122	0.6325
2017	0.5869	0.6258	0.6211	0.6350	0.6146	0.6285	0.6228	0.6146
2018	0.5907	0.6159	0.6374	0.6350	0.6211	0.6455	0.6252	0.6293
2019	0.5448	0.6149	0.6126	0.6181	0.6181	0.6108	0.6136	0.6117
2020	0.5465	0.5918	0.6009	0.5981	0.5991	0.6019	0.6120	0.5926
2021	0.5480	0.6039	0.6187	0.6276	0.6171	0.6228	0.6195	0.6179
2022	0.5811	0.5712	0.5951	0.6057	0.5813	0.6041	0.5862	0.5943
2023	0.5470	0.6244	0.6390	0.6309	0.6407	0.6366	0.6423	0.6285
2024	0.5452	0.6327	0.6490	0.6571	0.6596	0.6588	0.6457	0.6522
2025	0.5552	0.6407	0.6442	0.6328	0.6369	0.6539	0.6279	0.6466
mean	0.5657	0.6159	0.6265	0.6292	0.6234	0.6316	0.6240	0.6236

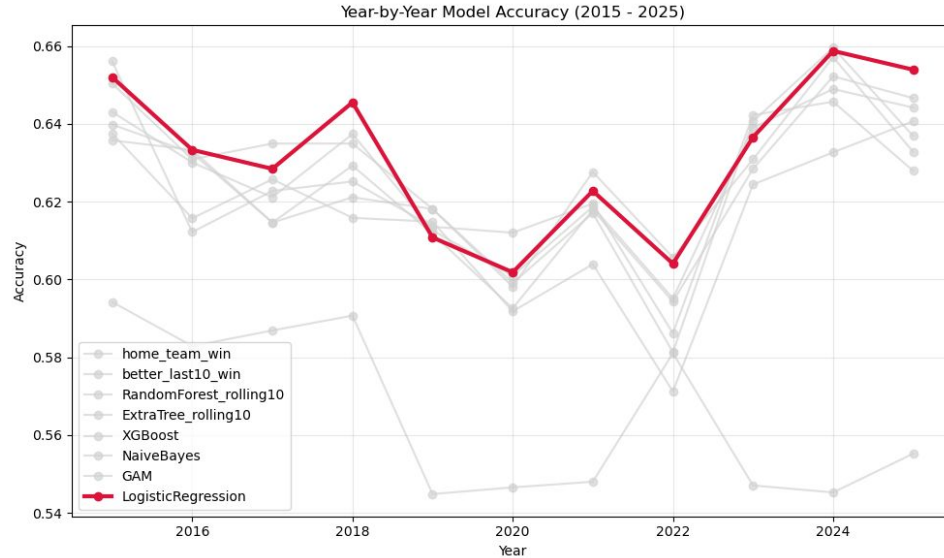
Year-by-Year Model Accuracy (2015 - 2025)



Year-by-Year Model Accuracy (2015 - 2025)



Model Selection



Paired comparisons vs. LogisticRegression_ewm:

	mean_diff	BEST bests (out of 11)	t_stat	p_value
model				
home_team_win	0.0660	11	8.2947	0.0000
better_last10_win	0.0158	10	4.7357	0.0008
NaiveBayes_ewm	0.0126	10	3.5559	0.0052
GAM_ewm	0.0106	9	3.4615	0.0061
XGBoost_ewm	0.0112	8	2.9155	0.0154
ExtraTree_ewm	0.0053	7	2.1100	0.0610
RandomForest_ewm	0.0045	8	1.7599	0.1089

- Logistic Regression is consistently at or near the top.
- Its advantage over tree-based ensembles (RF/Extra Trees) is not of statistical significance.
- Our models performed better on regular season games than on regular + playoff games

Conclusions

- **The 65% Performance Ceiling:** Models converged at **63%–66% accuracy** (ROC ~ 0.67). This represents the irreducible "any given night" noise floor of the NBA, not a modeling failure.
- **Significant Edge:** Delivered a **10%+ performance jump** over the naive Home Advantage baseline (55.4%).
- **Robust Signals:** Consistent results across Logistic Regression, GAMs, and Ensembles prove the predictive signal is stable and primarily linear.

Future works

- **Player-Level Tracking:** Shift from team aggregates to individual player efficiency, usage, and matchup metrics.
- **Live Injury Integration:** Automate pipelines to dynamically adjust team strength for injuries and load management.
- **Lineup & Postseason Optimization:** Model 5-man lineups and build dedicated playoff models to handle high postseason variance